

WEB FULL STACK WITH NODEJS



React JS is the most popular JavaScript framework and platform for developing client-side (front-end) mobile and desktop web apps or single page applications (SPAs).

HTML

INTRODUCTION TO HTML

- headings
- paragraph
- formatting tags
- image
- anchor
- lists
- table
- ID
- class
- iframe
- div
- span
- form
- audio
- video
- html vs html5

CSS

Introduction to CSS

ways to use CSS (inline, internal, external)

css selectors

css color

border

padding

margin

height

width

lists

text

display (none, inline, block, inline-block)

flex

grid

position

z-index

media

Java Script

Js Basics

- keywords
- identifiers
- datatypes
- variables and constants
- tokens
- operators

Decision Making Statements

Conditions Statements Or Selection Statements

- if
- if else
- nested if else
- if else if ladder
- switch case

Iteration Statements Or Loops

- for
- while
- do while
- for in
- for of

Jump Statements

- continue
- break

Functions

- function declarations, definitions and calling
- types of functions
- anonymous functions
- arrow functions or arrow functions
- self invoked function (IIFE)
- storage classification or variable scope (var let const and without keywords)
- recursion

Strings Methods

array Methods

Global Execution Context and Function execution context

- scope chain rule and lexical scoping
- closure

Objects

- literal and keys
- object functions
- nested objects
- Object Prototypes
- setter and getter methods

Set and Map

WeakMap and WeakSet

Dates and Time

OOPS

- literal and keys
- object functions

- nested objects
- Object Prototypes
- setter and getter methods

DOM Manipulation

- DOM
- BOM
- event handling
- time based events

Advanced Js

How Javascript Works

hoisting

strict mode

closure

synchronous and asynchronous programming

Event propagation

bubbling

capturing

Event loop

function curring

ajax and json

calling api using ajax

callback hell

promise

async await

fetch

modules

version

Typescript

- Why Typescript
- Basic Types
- Class and Interfaces
- Modules

React Js

INTRODUCTION TO REACT JS

- What is React JS?
- What is SPA?
- DOM vs Virtual DOM
- Advantages and Disadvantages
- Key Features

ENVIRONMENTAL SETUP

- Node | NPM
- Installation of CLI
- Setup Project
- Directory Structure
- Code Editors
- How React JS Application Boot

BASIC FEATURES OF REACT JS

- React Concepts
- JSX and TSX

- Render Elements
- Function and Class Components
- Props and State
- Handling Events
- Dynamic Data Rendering
- Property Binding

KEY FEATURES OF REACT JS

- Conditional Rendering
- List and Keys
- Forms Handling
- Forms Validations

COMPONENT LIFECYCLE HOOK

- Understanding component lifecycle
- All Lifecycle Hooks

EVENT HANDLING REACT

- Understanding React Event System
- Passing arguments to event Handlers

NETWORK CALL

- Fetch
- Axios

CUSTOM SERVICES

- Introduction to Services
- Building a Service

LOCAL DATA STORAGE

- Local Storage
- Session Storage
- Cookies

ROUTING WITH REACT ROUTER

- Setting up React Router
- Configuring route with Route Component
- Making routes dynamic with Route Params
- Working with nested routes
- Link and NavLink
- Redirect Routes

UI COMPONENTS

- Material Design
- PrimeNG

INTRODUCTION TO REDUX

- Why Redux
- Install and setup
- Store, Reducer, actions
- Dispatcher
- High order Components
- mapStateToProps and map DispatchToProps usage

ADVANCE REDUX

- Async Actions
- Middleware
- Redux Thunk and Redux Saga

React Hooks

- Why We Need Hooks.
- Different Types Of Hooks

- Using State And Effect Hooks
- Usereducer, Useref Etc.
- Custom Hooks
- Rules Of Hooks

Third Party Modules

- Social Login
- Pagination
- Search
- Filter
- JWT Token
- File Upload
- Many More

Rest Js Testing

- Jest with Enzyme

Develop a CRUD Application in React Js

Node Js

Introduction to Node.js

- What is node.js
- A word on event loop
- Event driven architecture
- Blocking vs non-blocking code
- Node Process Model
- Setup node.js
- Node REPL
- Debugging node.js app

Modules in JS

- Understanding exports and require
- Creating modules
- Importing modules
- Quick tour on NPM
- Installing 3rd party modules

Events & streams

- Significance of Events
- Event Emitter class
- Emitting and listening to events
- Types of streams
- Working with streams
- Composing streams using pipe

Working with Process and File System

- Process object
- Handling exceptions at process level
- Listening to process events
- Working with file system using fs module
- Creating, Copying, Deleting files
- Creating, Copying, Deleting directories
- Watching for file changes

HTTP

- Building a Web Server
- HTTP Request Methods, Headers
- Response Codes and Headers

Express Js

Expressjs Introduction & Setup

- Building a Web Server
- HTTP Request Methods, Headers
- Response Codes and Headers
- Serving static files with express
- Express application generator

Routing

- Route Methods
- Route Paths
- Parameterized Routes
- Route Handlers
- Express Router

Middleware

- Types of middleware
 1. Application
 2. Router
 3. Error
 4. Built-in
 5. Third-party
- Error middleware
- Body parser, cors, cookie parser, session management

Template Engines

- What are template engines
- EJS
- Jade
- Handlebars

Expressjs security

1. Authentication
 2. JWT
 3. Securing routes
- Debugging in Express
 - Realtime programming with Socket.io

Mongo DB

Introduction to MongoDB

- Installing MongoDB
- The current SQL/NoSQL landscape
- Document-oriented vs. other types of storage
- Mongo's featureset

- Common use-cases
- MongoDB databases
- MongoDB Collections
- MongoDB Documents

CRUD Operations in MongoDB

- Creating documents
 1. insert()
 2. update()
 3. save()
- Querying documents
 4. find()
 5. Working with equality
 6. Query operators
 7. Building complex queries
- Updating documents
- Deleting documents
- Limit
- Order By

Introduction to Mongoose

- Word on ORM/ODM
- Installing mongoose
- Connecting to MongoDB from mongoose

Core concepts of Mongoose

- Understanding mongoose schemas and datatypes
- Working with Models
- Using modifiers in schema
- Using virtual fields
- Optimizing query performance by enabling indexes

Mern Application Deployment

- Build Application
- Deploy in Heroku

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